

Mathematics Practice Questions

- 1) If $\det(A) = 0$, then the matrix A is:- A) Orthogonal B) Symmetric C) Invertible D) Singular
- 2) The determinant of a triangular matrix is: A) Product of diagonal elements B) Sum of diagonal elements
C) Zero D) One
- 3) The pair of equations $x+2y-5 = 0$ and $-4x-8y+20 = 0$ have : A) Unique solution B) Exactly two solutions
C) Infinite many solutions D) No solution
- 4) In how many ways a committee consisting of 5 men and 3 women, can be chosen from 9 men and 12 women ?
- 5) How many considerable words can be created by using 2 letters from the term "LOVE" ?
- 6) If $A = I$, then $A^2 = ?$
- 7) A bag contains 6 white and 4 black balls. Two balls are drawn at random from the bag. Find the probability that both the balls are of the same color ?
- 8) There are 7 purple clips and 5 brown clips. Two clips are selected one by one without replacement. Find the probability that the first is brown and the second is purple ?
- 9) If $P(A) = 7/13$, $P(B) = 9/13$ and $P(A \cap B) = 4/13$, find the value of $P(A|B)$.



- 10) If $(A+B)^2 = A^2 + 2AB + B^2$ then what can we say about A and B? (Assume AB and BA exists)
- 11) If A is a $n \times m$ matrix such that AB and BA are both defined, then order of B is:- ?
- 12) If $AB = A$ and $BA = B$, then show that A and B are idempotent matrices.
- 13) If A and B are n rowed squared matrices and $AB = 0$ & $|B| \neq 0$, then $A = 0$. State True or False.
- 14) Eigenvalues are obtained by solving: A) $\det(A) = 0$ B) $\det(A+I) = 0$ C) $\det(A-\lambda I) = 0$ D) $A = \lambda I$
- 15) A newly constructed bridge may fall down either due to wrong designing or by inferior material used in construction. The chance that the design is faulty is 10% and the probability of its collapse if the design is faulty is 95% and that due to bad material it is 45%. If the bridge collapses. Find the chance that it was due to wrong designing.
- 16) In a class, 40% of the students like Mathematics and 25% of students like Physics and 15% like both the subjects. One student select at random, find the probability that he likes Physics if it is known that he likes Mathematics.
- 17) The point where the Null Hypothesis gets rejected is called as ?
- 18) If $AB=BA$, then matrices A and B are said to be - ?
- 19) If $AB = 0$, does it imply that it is necessary that $BA = 0$.
- 20) What's the slope of the line $30x - 6y = 3$?
- 21) Two Notebook and one pen cost Rs. 35 and 3 Notebook and four pen cost Rs. 65. Find the cost of Notebook and pen separately.



22) Find the solution for the given pair of linear equations. - $3x + 2y = 10$ & $6x + 4y = 20$

23) Using Cayley–Hamilton theorem, every square matrix satisfies:

- A) Its determinant
- B) Its transpose
- C) Its characteristic equation
- D) Its inverse

24) Find the Eigenvalue for the given matrix. $A = \begin{bmatrix} 4 & 1 & \tilde{3} \\ 1 & 3 & 1 \\ 2 & 0 & 5 \end{bmatrix}$.

25) Find the Eigen vector for value of $\lambda = -2$ for the given matrix. $A = \begin{bmatrix} 3 & 5 \\ 3 & 1 \end{bmatrix}$.

26) Find the Eigen value and the Eigen Vector for the given matrix. $A = \begin{bmatrix} 3 & 4 & \tilde{2} \\ 1 & 6 & 2 \\ 1 & 4 & 4 \end{bmatrix}$

27) Find the value of $A^3 + 19A$, $A = \begin{bmatrix} 2 & -3 & 1 \\ 2 & 0 & -1 \\ 1 & 4 & 5 \end{bmatrix}$



28) Find the value of $A^3 - 3A^2 - 28A$, $A = \begin{bmatrix} -1 & 2 & 8 \\ -2 & 3 & 0 \\ -4 & 5 & 1 \end{bmatrix}$.

29) Let A and B be two events such that $P(A) = \frac{1}{5}$ While $P(A \text{ or } B) = \frac{1}{2}$. Let $P(B) = P$. For what values of P are A and B independent?

30) If A and B are two mutually exclusive events with $P(a) > 0$ and $P(b) > 0$ then it implies they are also independent.(TRUE/FALSE)

31) What is the probability of an impossible event?

32) If $A = A_1 \cup A_2 \dots \dots \cup A_n$, where $A_1 \dots A_n$ are mutually exclusive events then?

33) Company A produces 10% defective products, Company B produces 20% defective products and C produces 5% defective products. If choosing a company is an equally likely event, then find the probability that the product chosen is defective.

34) Suppose 5 men out of 100 men and 10 women out of 250 women are colour blind, then find the total probability of colour blind people. (Assume that both men and women are in equal numbers.)

35) A problem is given to 5 students P, Q, R, S, T. If the probability of solving the problem individually is $\frac{1}{2}$, $\frac{1}{3}$, $\frac{2}{3}$, $\frac{1}{5}$, $\frac{1}{6}$ respectively, then find the probability that the problem is solved ?



36) A jar contains 63 balls (1, 2, 3,....., 63). Two balls are picked at random from the jar one after one and without any replacement. What is the probability that the sum of both balls drawn is even?

37) What is the probability that the number is a perfect square?(up to 100)

38) The probability that the political party A does a particular work is 30% and the political party B doing the same work is 40%. Then find the probability that the work is completed if the probability of choosing the political party A is 40% and that of B is 60%.

39) In badminton practice session, the probability that the player A serves properly is 0.8 and that he player B serves properly is 0.9. If there are only two players, then find the probability that it is serves properly.

40) The probability that person A completes all the tasks assigned is 50% and that of person B is 20%. Find the probability that all the tasks are completed.?

41) Let there be two newly launched phones A and B. The probability that phone A has good battery life is 0.7 and the probability that phone B has good battery life is 0.8. Then find the probability that a phone has a good battery life.

42) Which of the following mentioned standard Probability density functions is applicable to discrete Random Variables?

a) Gaussian Distribution b) Poisson Distribution c) Rayleigh Distribution d) Exponential Distribution



43) The expected value of a discrete random variable 'x' is given by _____

44) Find which of the following is a Continuous random variable?

- a) Number of kids in a family
- b) Number of planets around the sun
- c) Number of tails tossing a coin four times
- d) Life of an electric fan

45) 5. Let X be the random a variable, $P(X=x)$ is the Probability mass function is given by

X	0	1	2	3
$P(X=x)$	0	$1/2$	$2k$	$3k$

Determine the value of k ?

46) Determine the value c so that the following function can serve as a probability distribution of the discrete random variable x : $f(x)=c(x+4)$, for $x=0,1,2,3$

47) A dice is thrown, what is the probability of getting an odd number?

48) A dice is thrown, what is the probability of getting multiples of 3?

49) What is the probability of picking an ace from a pack of cards?

50) What is the probability of picking a card of the club from a pack of cards?

